Lannik Psi: expected range performance

October 2026 · before the antenna supplier's design data · Patrik Andersson

Internal. This page quotes figures from Infineon's CTRX8188F datasheet, which we hold under NDA.

Lannik Psi is our long-range, narrow-beam 76–77 GHz radar, built on Infineon's CTRX8188F radar chip with eight transmit (TX) and eight receive (RX) channels. This page gives our current estimate of how far it detects a small target, how that compares with what we said in June, and how far we can trust it now that we have measured the same chip on Infineon's evaluation radar, CARKIT. The antenna design is with a supplier. Their design data will replace some of the assumptions here, and the estimate will be updated when it arrives.

In brief

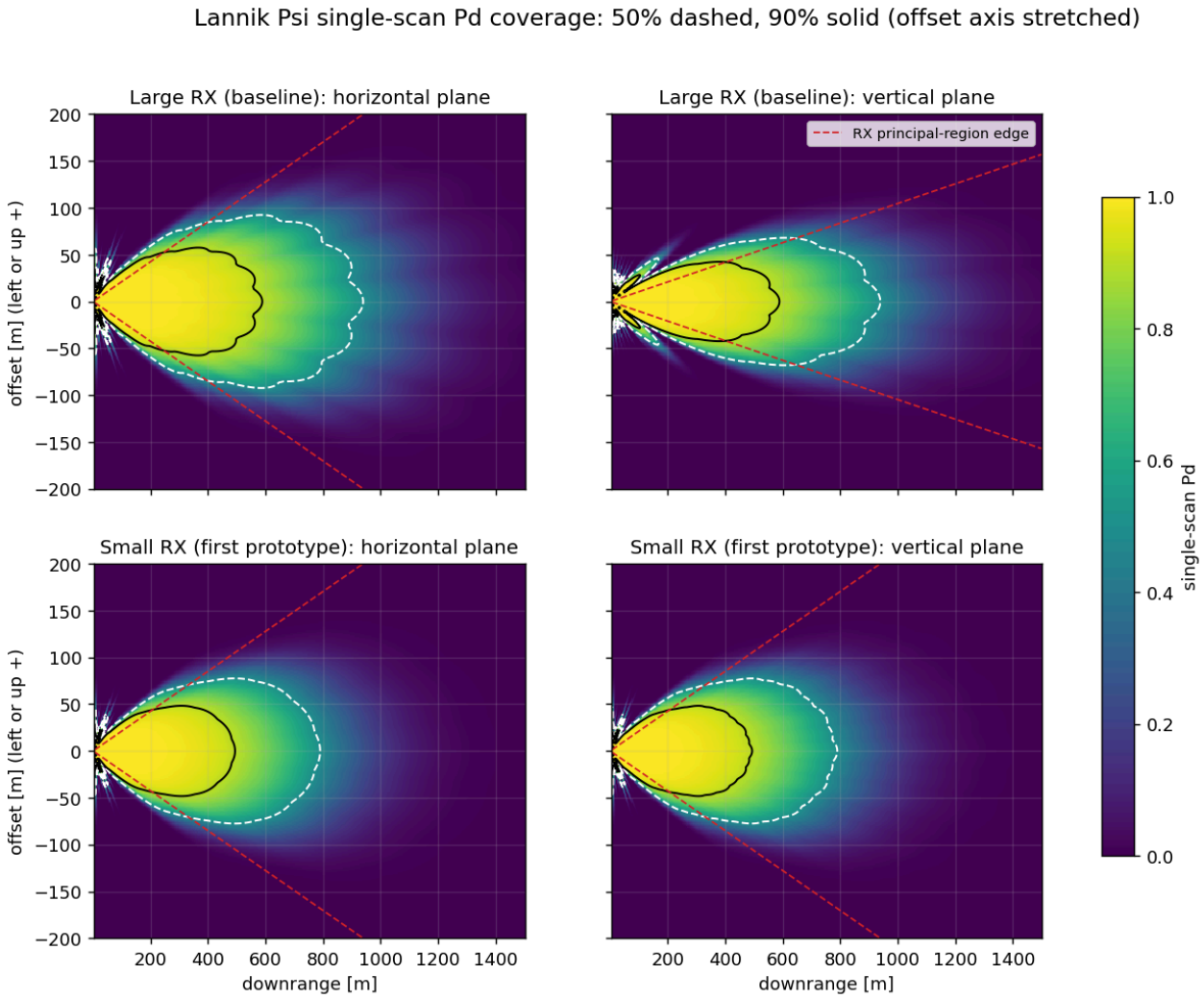
- **About 940 m straight ahead.** With the larger of the two receive antennas under development, a target of 1 m² radar cross-section is detected in a single frame with 50 % probability out to 939 m on boresight, and with 90 % out to 588 m. The smaller receive antenna, ordered first as the prototype, reaches 789 m. Six degrees off boresight the ranges are 643–766 m.
- **About 5 % less than the June estimate of 994 m.** June used placeholder antennas and no losses. Real antenna designs, a stricter false-alarm budget, and the antenna and atmospheric losses brought the estimate to 913 m by the end of the antenna study. Running the receiver at its +3 dB gain setting, which is now settled, adds 26 m.
- **The model holds up against measurements.** On CARKIT, the radar equation with the chip's datasheet values came out within 1 dB of a measurement against a reference reflector, about 5 % in range.
- **The main uncertainties are the antenna's losses, the radome and the spread between chips.** The antenna losses here are the design targets we gave the supplier. Each 0.1 dB of radome loss, each way, costs about 1 % of range, and a chip at the datasheet's worst-case noise figure about 15 %.
- **Two design rules came out of the measurements.** The radar must leave the chip's synthesizer enough time between chirps, and it must set the receiver gain to +3 dB explicitly.

What the numbers mean

- **Single-scan detection probability (Pd)** is the probability of detecting the target in one frame. We quote the range where it reaches 50 %, which a tracker can follow, and 90 % for comparison.
- **The target** is 1 m² and fluctuates from frame to frame (Swerling 1).
- **False alarms** are held at one in a million per range–Doppler cell over all receive beams. With 512 range and 512 Doppler cells per frame, that is about 0.26 false alarms per frame, or about 5 per second at an assumed 20 frames per second. It is a placeholder until the false-alarm rate the tracker can accept is set.
- **The radar model** takes the chip's typical datasheet values (14.5 dBm TX power per channel, 9.7 dB receiver noise figure at +3 dB gain), ideal antenna models with 1.0 dB loss on each side (the targets in our request for quotation), the standard atmosphere at 1000 m altitude, the lowest altitude of the customer's use cases, and a small loss for the chip's phase noise. It leaves out the radome, rain and clutter.

We do not quote the probability of acquiring a track. It depends on the confirmation rule, the frame rate and how the target approaches, none of which is defined yet. A confirmed track forms further out than single-scan Pd suggests, so single-scan Pd is the conservative measure.

Coverage

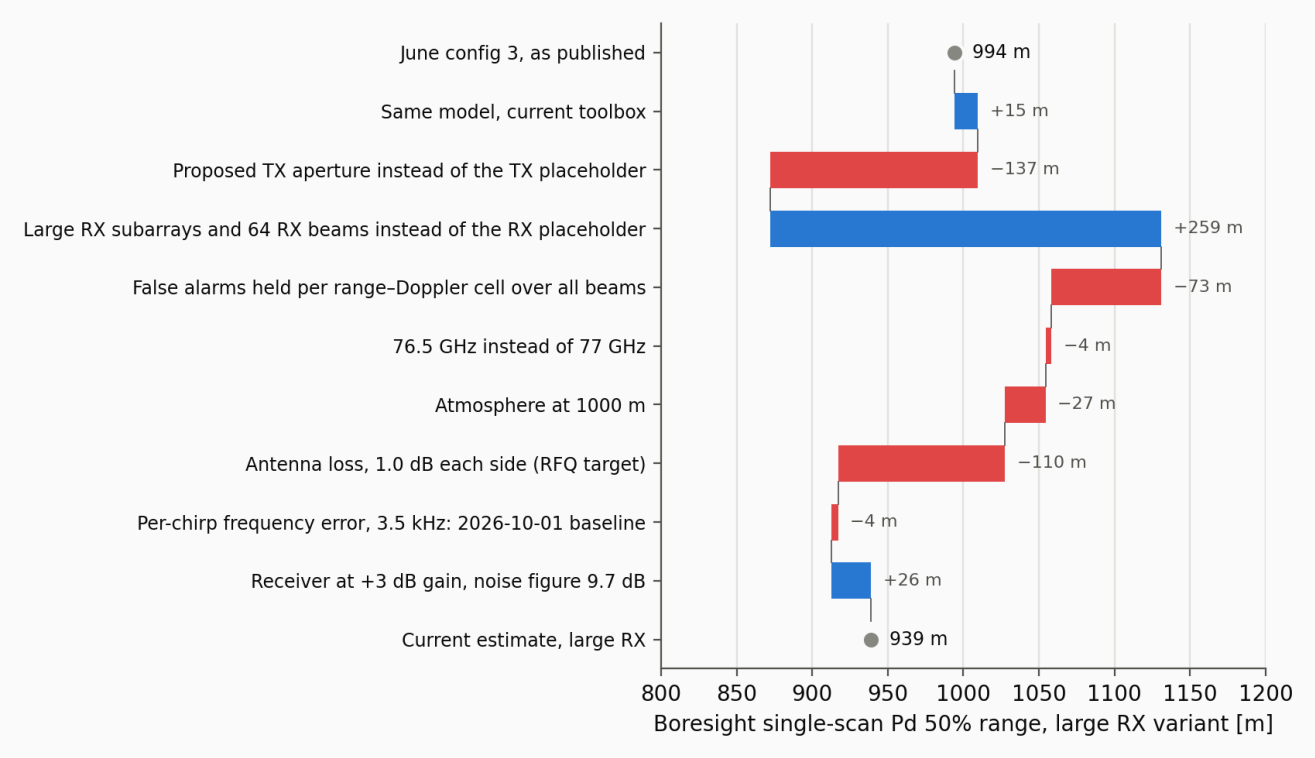


Probability of detecting a 1 m² target in one frame, by position, for the large receive antenna (top) and the small one (bottom), in the horizontal (left) and vertical (right) planes. The dashed contour is 50 %, the solid one 90 %. The red dashed lines mark the edges of the region the receive array resolves without ambiguity.

Direction	Large RX: Pd 50% / 90%	Small RX (first prototype): Pd 50% / 90%
Boresight	939 / 588 m	789 / 493 m
6° horizontally	766 / 478 m	643 / 401 m
6° vertically	646 / 403 m	643 / 401 m

The beam is narrow by design: it trades width for gain. Six degrees is about the half-width of the TX beam. The large variant’s taller receive antennas narrow its vertical coverage to the small variant’s at 6°.

Compared with June



Each bar is one change to the model, in the order they were made, applied to the bar above. Blue lengthens the range, red shortens it.

The June configuration comparison estimated 994 m on boresight for this configuration with 17 dBi placeholder antennas, no losses and a false-alarm probability per receive beam. Since then:

- **The antennas became real designs.** The proposed TX antenna widens its beam on purpose, which costs 2.5 dB against the placeholder. The large receive antennas more than make up for it, by 4.5 dB.
- **False alarms are now counted over all 64 receive beams** rather than per beam, which costs 1.2 dB.
- **Losses are now explicit:** 1.0 dB of antenna loss on each side, the atmosphere at 1000 m and the move to 76.5 GHz, together 2.5 dB.
- **The receiver runs at +3 dB gain**, whose noise figure is 0.5 dB lower.

Step	Pd 50% / 90%	Change in Pd 50% range	Equivalent SNR
June config 3, as published	994 / 614 m		
Same model, current toolbox	1009 / 623 m	+1.6%	+0.27 dB
Proposed TX antenna instead of the placeholder	872 / 538 m	−13.6%	−2.54 dB
Large RX antennas and 64 receive beams instead of the placeholder	1131 / 698 m	+29.7%	+4.51 dB
False alarms per range–Doppler cell over all beams	1058 / 655 m	−6.4%	−1.15 dB
76.5 GHz instead of 77 GHz	1054 / 653 m	−0.4%	−0.06 dB
Atmosphere at 1000 m	1028 / 642 m	−2.6%	−0.45 dB
Antenna loss, 1.0 dB each side	917 / 573 m	−10.7%	−1.97 dB
Phase-noise loss, 3.5 kHz per-chirp error: end of the antenna study	913 / 572 m	−0.5%	−0.09 dB
Receiver at +3 dB gain, noise figure 9.7 dB: **current estimate**	**939 / 588 m**	+2.9%	+0.49 dB
Small RX antenna instead, same assumptions	789 / 493 m	−16.0%	−3.02 dB

The equivalent SNR is the change in signal-to-noise ratio that would move the range as much under the radar equation's fourth-power law, 40 $\log_{10}$ of the range ratio. In total the current estimate is 5.5 % below June's, 0.99 dB. The antenna study's own table differs from this one by a metre or two in some rows, from later refinements of the false-alarm calculation.

What the CARKIT measurements add

We measured CARKIT, Infineon's evaluation radar with the same chip, against our model from September to October (the CARKIT validation report):

- **The radar equation with datasheet values is right to within about 1 dB.** With a reference reflector on a fixed mount in a measurement chamber, CARKIT came out 0.4–0.7 dB below the model. CARKIT has a different antenna and its own radome, so this confirms the chip and the model, not Psi's antenna.
- **The receiver's +3 dB gain setting behaves as the datasheet says.** The chamber result above was at +3 dB with the datasheet's 9.7 dB noise figure, and switching to 0 dB changed the receiver noise as the datasheet predicts. Infineon confirmed that the datasheet's two noise modes are these gain steps. Psi will run at +3 dB, so the estimate now uses 9.7 dB instead of the chip default's 10.2 dB.
- **The phase-noise allowance holds if the chirps are timed with margin.** With a 10 µs ramp, the chip's synthesizer needed about 15–25 µs between chirps to settle, far more than its datasheet suggests. With that, the per-chirp frequency error was 3.1–3.6 kHz, about the 3.5 kHz assumed here, and Psi's longer ramp should average it down further. It costs about 0.1 dB at 1 km. With Infineon's original chirp timing it was 14–30 kHz, which would cost 1.5–7 dB at 1 km. The sensitivity table shows what 21 kHz would cost.
- **Some effects matter for the product but are not in these numbers.** Strong objects close to the radar raise the noise floor at all ranges through the chip's phase noise, by an amount that depends on the installation. Near traffic, other radars interfered in up to a fifth of the frames. The receiver is slightly noisier at the lowest frequencies, which correspond to short ranges, but not at the 500–800 m of the use cases.

Sensitivities

Each row changes one assumption of the large variant's estimate:

Change	Pd 50% range	Change	Equivalent SNR
Noise figure at the datasheet maximum, 12.7 dB	794 m	−15.5%	−2.9 dB
Per-chirp frequency error of 21 kHz (too little time between chirps)	828 m	−11.8%	−2.2 dB
Light rain, 1 mm/h (attenuation only)	844 m	−10.1%	−1.8 dB
TX power 1 dB lower (temperature)	888 m	−5.4%	−1.0 dB
Receiver left at the chip's default 0 dB gain	913 m	−2.8%	−0.5 dB
Sea-level instead of 1000 m atmosphere	926 m	−1.4%	−0.2 dB
Radome, 0.2 dB one way	918 m	−2.2%	−0.4 dB
Radome, 0.5 dB one way	888 m	−5.4%	−1.0 dB
Radome, 1.0 dB one way	839 m	−10.6%	−2.0 dB

The radome is not yet defined, so these are examples, not a typical value. It acts on both passes: each 0.1 dB each way costs 1.1 % of range. Losses add in dB, so a radome of 0.5 dB each way and 1 dB less TX power together cost as much as a 1.0 dB radome, 10.6 %. The phase-noise loss and rain grow with range, so they cost less at the Pd 90% range: 5.6 % and 6.8 % there.

Relation to the customer use case

The customer's first use case asks for detection of a 1 m² target at 500–800 m within ±8°, without stating a detection probability, false-alarm rate or confirmation rule; the use cases are indicative and subject to revision. The estimate reaches 939 m on boresight and 646–766 m at 6°. The design

deliberately accepts a beam narrower than ±8° for more gain. The use case also puts the radar behind a radome, which these numbers leave out (see the sensitivities).

Not yet established

- **The antenna's real performance.** The supplier's predicted gain and patterns will replace the ideal models and the 1.0 dB loss targets.
- **The radome**, which the product needs and which is not yet defined.
- **The waveform**, including chirp timing with margin for the synthesizer.
- **The signal-processing chain in detail.** The estimate assumes standard processing: windowed range and Doppler transforms, a cell-averaging detection threshold and the best of the 64 receive beams. The chain's actual design can change some of the numbers.
- **Requirements:** detection probability, false-alarm rate and closing speeds for the use cases.
- **What the model leaves out:** clutter, interference, strong returns near the radar, and tracking.

Documents and reproducing

File	Contents
psi_performance.py	Computes every number and figure here from the antenna study's model, with the receiver at +3 dB
generated/summary.json	The numbers
Lannik Psi antenna-concept study	The antenna concept, the model and its assumptions, concluded 2026-10-01
CARKIT validation study	The measurements behind the model's confidence and the design rules
June configuration comparison	The June estimate

From the repository root, `make study_261010_lannik_psi_performance` runs the script in a few seconds.